

Unit 4: Data Preprocessing

Data Splitting – Class Activity Sheet #9

Holdout Splitting Stratification Validation Sets Cross Validation

Name		Batch	
Date		Global ID	

Learning outcome :

After completing this activity, you should be able to select an appropriate splitting method, explain the purpose of training, validation, and test sets, implement stratified splitting, and apply cross-validation without data leakage.

Part A Quick Concept Check

No	Statement	Your answer
1	The test set may be used repeatedly to choose hyperparameters.	True / False
2	Stratification preserves the target-class proportion across subsets.	True / False
3	In five-fold cross-validation, every sample is used for validation once.	True / False
4	Time-series data should usually be randomly shuffled before splitting.	True / False
5	SMOTE should be applied before the train-test split.	True / False

Explain why the test set must remain untouched until final evaluation.

Part B Choose the Correct Splitting Method

Choose from: random holdout, stratified holdout, group split, time-series split, k-fold cross-validation, or stratified k-fold cross-validation.

Scenario	Recommended method	Reason
Fraud dataset with 2% fraudulent transactions		
Daily sales records ordered from 2023 to 2026		
Several scans are available for each patient		
Small balanced tabular dataset		
Large image dataset requiring repeated tuning		

Part C: Two Way and Three Way Splitting

A dataset contains 2,000 records. Complete the table. Show your calculations.

Split	Training	Validation	Testing
80:20 two-way	1600	Not used	400
70:15:15 three-way	1400	300	300
60:20:20 three-way	1200	400	400

When is a separate validation set preferable to cross-validation?

Part D: Stratified Splitting by Calculation

A medical dataset contains 1,000 patients: 900 Negative and 100 Positive. Apply an 80:20 stratified train-test split.

Subset	Total records	Negative -900	Positive-100	Positive percentage
Training	800	720	80	
Testing	200	180	20	

What could happen if an ordinary random split placed very few positive cases in the test set?

Part E Python Practice

Use the sample classification dataset below. First predict the class distribution, and then run the code to verify it.

```
import pandas as pd
from sklearn.model_selection import train_test_split

df = pd.DataFrame({
    'Hours': [1,2,2,3,3,4,4,5,5,6,6,7,7,8,8,9,9,10,10,11],
    'Attendance': [50,55,58,60,62,65,68,70,72,75,76,78,80,82,84,86,88,90,92,95],
    'Pass': [0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,1,1,1,1]
})
```

1. How many records belong to each class?
2. Task 1 Write the code to perform a random 75:25 split. Set random_state=42. Record the class counts in y_train and y_test.

Subset	Class 0 count	Class 1 count
Training		
Testing		

3. Task 2 Repeat the split using stratify=y. Compare the result.

Subset	Class 0 count	Class 1 count
Training		
Testing		

Which split is more representative of the full dataset? Justify your answer.

Part F Create a Three Way Split

Complete the missing arguments to create a 70:15:15 split. Remember that the second split divides the 30% temporary set into two equal parts.

```
X_train, X_temp, y_train, y_temp = train_test_split(
    X, y, test_size=____, random_state=42, stratify=y
)

X_val, X_test, y_val, y_test = train_test_split(
    X_temp, y_temp, test_size=____, random_state=42, stratify=y_temp
)
```

Subset	Expected percentage	Purpose
Training	70%	
Validation	15%	
Testing	15%	

Part G Cross Validation by Hand

Ten observations are divided into five folds. Each fold contains two observations. Complete the validation fold and training folds for each iteration.

Iteration	Validation fold	Training folds
1	Fold 1	
2		
3		
4		
5		

The five validation accuracy scores are 0.80, 0.85, 0.75, 0.90, and 0.80.

Calculate the mean cross-validation accuracy.
Why is the mean score generally more reliable than the result of one train-validation split?

Part H Implement Cross Validation

```
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import StratifiedKFold, cross_val_score
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler

model = make_pipeline(StandardScaler(), LogisticRegression())
skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
scores = cross_val_score(model, X, y, cv=skf, scoring='accuracy')

print('Fold scores:', scores)
print('Mean:', scores.mean())
print('Standard deviation:', scores.std())
```

Record the fold scores, mean, and standard deviation.

What does a large standard deviation across folds suggest?

Part I Detect Data Leakage

For each workflow, write Correct or Incorrect. If incorrect, rewrite the sequence.

Workflow	Decision	Corrected sequence
Scale the complete dataset, then split it.		
Split data, fit the scaler on training data, transform validation and test data.		
Apply SMOTE to all data, then create five folds.		
Keep a final test set aside and run cross-validation only on training data.		
Tune hyperparameters using test accuracy.		